

AUSTIN GIFFEN

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2024 U.S. Presidential Scholar, National Merit Scholar

EDUCATION

Yale University (GPA: 3.91)

New Haven, CT

Major: Electrical Engineering

Graduating Fall 2027

- **Relevant Courses**

- Probabilistic Machine Learning, Quantum Information Processing and Communication, Mechatronics Laboratory, Circuits and Systems Design, Digital Systems, Information Systems, Intensive Physics (classical mechanics/special relativity), Modern Physical Measurement, Linear Algebra, Computer Engineering, Intensive Physics (E&M/special relativity/quantum mechanics), Electromagnetic Waves and Devices, Electronics, Communications and Control

WORK EXPERIENCE

Avishtech: Electrical Engineering Intern

- **Built analytical via in cavity models that match full wave model accuracy in orders of magnitude less time up to 40GHz**
- **Built models in Sonnet and OpenEMS to validate analytical model for a variety of canonical structures.**

Yale Undergraduate Research Assistant — Konezny Lab

January

2025 - January 2026

- Recipient of the Yale College First-Year Summer Research Fellowship (\$5,000 research grant)
- Utilized semiconductor principles to evaluate and optimize the conduction mechanisms of CuSCN to improve its potential as a hole transport layer material in solar cells.
- Designed and assembled gas flow control system using Arduino, mass flow controllers, and LabVIEW for precise hydrogen/oxygen doping experiments.
- Developed Python scripts to process experimental data, compute conductivity, and generate visual correlations.
- Reviewed and applied findings from academic literature to refine experimental design and interpret discrepancies in conductivity behavior.

https://docs.google.com/presentation/d/1e6wMXEWEJpyx848cdU-yOOiMJCMuljVILvgJ71IJeLk/edit?slide=id.g28067cb0407_0_1#slide=id.g28067cb0407_0_1

Yale Undergraduate Research Assistant — Jung Han Lab

January 2026 - Present

- Developed and optimized machine-learning models (Random Forest, cross-validated grid search) in Python to predict glucose levels from optical spectroscopy sensor signals.

Lead Sailing Instructor for Southern Idaho Sailing Association

May

2022 - August 2024

- Worked with two other instructors to run sailing camps for beginner through advanced sailors.

Soccer Referee

January

2022 - June 2024

- NFHS, SWISRA, SISL certified – Officiated DPL League, ECNL League, and state tournaments.

PROJECTS

- **Yale CubeSat (satellite) Cosmic Ray Detector Subteam Co-Lead**
September 2024 - Present
 - Coordinate with NASA regarding launch plans and progress, Lead and coordinated subteam projects, collaborated with other subsystem leads, and supported outreach initiatives
 - Developed the complete cosmic ray detector schematic and PCB from negative voltage rail, multiple stage amplifiers, comparators, DACs, counter ICs, temperature sensors, boost converters, and multiplexers.
 - Designed a precision negative voltage rail PCB using a 555 timer IC (NE555) and operational amplifier.
 - Performed component selection and bill of materials optimization.
 - Simulated all circuits in KiCad using SPICE models for each component to validate gain, noise, timing response, and power behavior before fabrication.
 - Tested and validated the detector subsystem through bench measurements, debugging, calibration, and data analysis from local cosmic-ray counts to confirm proper functionality.
- **Mind Vault App**
 - Engineered a full-stack learning ecosystem leveraging React, TypeScript, Node.js/Express, and Firebase Firestore inside a pnpm monorepo to convert AI responses and video content into active-recall quizzes.
 - Developed a Manifest V3 Chrome Extension (React/Vite) that injects context-aware capture triggers across ChatGPT, Gemini, Claude, and YouTube for seamless content capture.
 - Architected shared core libraries (quiz-engine) to unify flashcard evaluation, streak management, and gamification logic across both the web client and browser extension.
- **AnalogLab Platform:**
 - **Unified Developer Platform:** Built an end-to-end platform enabling developers to sample data, train, and deploy edge compute models onto diverse analog AI backends.
 - **Broad Architecture Coverage:** Designed general-purpose model translation pipelines supporting Continuous-Time Analog, Compute-In-Memory (CIM) crossbars, and Event-Driven Neuromorphic chips—covering 95% of analog edge silicon.
 - **Automated SDK Interoperability:** Standardized output files to compile seamlessly into chip-specific SDK toolchains, eliminating hardware-dependent code rewrites.
- **Developed a Hardware-Aware Software Layer (HASL) compiler framework to translate Bayesian Neural Networks (BNNs) into physically realizable analog memristive crossbar circuit netlists for uncertainty-aware edge computing.**
2026 Spring Semester
 - **Optimized Variational Training:** Formulated a hardware-penalized Evidence Lower Bound (ELBO_{γ}) incorporating experimental memristor constraints ($\sigma = h(\mu)$), halving the parameter search space by expressing weight variance directly as a function of mean conductance.
 - **Minimized Routing Architecture Costs:** Mathematically proved that memristive crossbar topology achieves the theoretical minimum allocation cost ($\text{Cost}(\mathcal{A}_{\text{crossbar}}) = \sum_{n=1}^N n \cdot n_I$) for forward-pass probabilistic inference by minimizing inter-block data movement.
 - **Derived Sampling Convergence Guarantees:** Extended the compiled hardware substrate into an analog iterative sampler using Equilibrium Propagation (EP) and the Unadjusted Langevin Algorithm (ULA), establishing explicit KL-divergence convergence bounds under physical device noise.
- **Automated AI Video Generation Pipeline (Docker, CometAPI, Sora-2/Gemini)**
Summer 2025
 - Built and deployed a fully automated, Dockerized video generation system on a headless server, integrating CometAPI (Sora-2 / Gemini Flash Lite) to generate short-form videos from dynamically created prompts.
 - Implemented end-to-end automation including prompt generation from trending YouTube data,

scheduled execution, secure credential management, OAuth token refresh, and programmatic YouTube uploads via the YouTube Data API.